Dilemma Dynamics

Cooperation, deliberation and repeated-game incentives

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Evaluation and mathematical notes

Abstract

Does drafting before acting help an AI recognise when cooperation pays? We compare Direct and bounded Draft procedures in Nemotron Nano and Super across 7,104 repeated Prisoner's Dilemma episodes at temperatures 0 and 1. Exact payoff references against disclosed policies separate decision quality from cooperation between autonomous agents. Draft reduces average benchmark loss by 16.8–37.5% across four planned comparisons, yet increases loss against Tit-for-Tat and reduces self-play cooperation. These results connect individual payoff decisions to the dynamics of reciprocity, and motivate experiments on contingent planning, relationship repair and richer interactions.

1 Motivation and overview

AI agents may interact repeatedly with people or other agents while pursuing their own objectives. Their choices can influence whether others cooperate, retaliate or rebuild a productive relationship. Understanding how individual decision quality relates to cooperation is therefore useful when evaluating agents that interact over time.

Repeated Prisoner's Dilemma makes this question concrete: immediate rewards compete with future consequences. Dilemma Dynamics combines exact payoff comparisons against disclosed opponent policies with autonomous pairings that reveal how cooperation develops, breaks down and recovers. Researchers and agent developers can inspect responses to incentives; students can explore the same ideas through play, saved recordings and mathematical explanations.

Intellectual setting. Rasmusen's account of repeated games, commitments and best responses supplies the theoretical foundation [1]. Akata et al. study social chain-of-thought in repeated games [2]; Fontana et al. show model-specific responses to opponent hostility [3]; Pal et al. examine strategies across histories and game settings [4]. Taylor and Bergen connect reasoning and action–message incongruence in communication games [5]. Together, these works motivate separating payoff competence, cooperation and communication. Here the scientific protocol uses sealed actions and asks how a bounded draft changes decisions and interaction.

2 Game, procedures and evaluation

2.1 Incentives and the two procedures

Players maximise expected own cumulative points. Both choose privately from cooperate (C) and defect (D), then reveal together. The matrix lists (own points, other points):

Own action	Other cooperates	Other defects
Cooperate	$(3, 3)$	$(0, 5)$
Defect	$(5, 0)$	$(1, 1)$

Thus temptation $T = 5$, reward $R = 3$, punishment $P = 1$ and sucker's payoff $S = 0$ satisfy $T > R > P > S$. Defection pays more in one round; a responsive partner can make cooperation valuable over repeated encounters.

An *episode* is one complete interaction. A fixed horizon announces its length H; a geometric horizon continues independently after each round with probability δ, giving $\mathbb{E}[N] = 1/(1-\delta)$ rounds and no announced last round. For example, $\delta = .9$ gives ten expected rounds, a different information setting from ten announced rounds.

The hosted models are NVIDIA Nemotron Nano and Super [6, 7, 8]. *Direct* makes one thinking-off action request, capped at 128 completion tokens. *Draft* makes a thinking-on draft capped at 1,024 tokens, followed by a separate thinking-off action request capped at 128 tokens. Both receive completed history; only the current draft is added to its action request. An *arm* is a model–procedure combination. Each episode uses temperature $\tau \in \{0, 1\}$ throughout, with top-$p = 1$. The comparison concerns the whole two-stage procedure, including its added computation and conditioning.

2.2 Two complementary tracks

Track A: disclosed opponents. Always Cooperate (AllC), Always Defect (AllD), Tit-for-Tat (TFT: C first, then copy the subject's preceding action), and Grim (C until a subject defection, then D forever) each face all three continuation probabilities .4, .6, .9. One-round AllC/AllD and announced ten-round TFT/Grim complete 16 conditions. This benchmark measures decisions against known rules.

Track B: autonomous interaction. All ten unordered pairings of the four arms, including self-play, interact at $\delta = .9$. Their response policies emerge during play. This track measures cooperation, joint return and breakdown/repair trajectories rather than applying the known-policy oracle.

The complete evaluation has **7,104 episodes**: 6,144 benchmark episodes and 960 autonomous dyads, with 3,552 episodes at each temperature. There are **48 paired blocks per temperature**, balanced across four neutral-action-label/display-seat strata with 12 blocks each. Every block has 16 benchmark conditions for each arm and ten autonomous pairings. Consequently, each arm has 768 benchmark episodes per temperature and each autonomous pairing has 48 dyads. Shared environment draws pair arms and temperatures; adjacent temperature pairs use frozen random order. Block count was selected using engineering resource forecasts before scientific outcomes.

2.3 Exact payoff loss and uncertainty

For a disclosed opponent, let s be its memory state, $r(s, a)$ the subject's payoff, and $f(s, a)$ the next state. Exact dynamic programming computes

$$Q^*(s, a) = r(s, a) + \delta V^*(f(s, a)), \qquad V^*(s) = \max_{a \in \{\mathrm{C,D}\}} Q^*(s, a). \tag{1}$$

Finite games use remaining-horizon values with $V_0^* = 0$ and backward induction. At a revealed decision, the gap $g_t = V^*(s_t) - Q^*(s_t, a_t) \geq 0$ measures the expected own payoff forgone by that action, allowing optimal play afterward. Episode loss is

$$L_{\text{geometric}} = (1-\delta)\sum_{t=1}^{N} g_t, \qquad L_{\text{fixed}} = \frac{1}{H}\sum_{t=1}^{H} g_t. \tag{2}$$

Zero means no achievable expected payoff was given up. Survival to a recorded round already supplies the geometric survival factor; multiplying each saved gap by δ^{t-1} again would count stopping twice. Loss has normalised own-payoff units, and is separate from the amount of cooperation.

Four primary Draft–Direct contrasts average the 16 conditions and four strata equally. Negative differences favour Draft. Uncertainty uses 50,000 deterministic stratified whole-block bootstrap draws. Approximate 98.75% marginal intervals apply the planned Bonferroni allocation of a 5%

family across four contrasts. The independent resampling unit is the matched block, not a round. Six planned temperature contrasts use nominal 95% intervals without multiplicity correction; opponent, presentation, planning and self-play diagnostics are exploratory.

3 Results and explanations

3.1 Drafting lowers average loss, with an opponent reversal

All four primary intervals lie below zero (Table 1). Relative loss reductions are 16.8% and 19.4% for Nano and Super at zero, and 25.7% and 37.5% at one. These are reductions in payoff forgone, not percentage increases in earned reward.

Model	τ	Direct	Draft	Draft–Direct	98.75% interval
Nano	0	.6462	.5374	$-.1088$	$[-.1640, -.0577]$
Super	0	.4616	.3720	$-.0896$	$[-.1212, -.0591]$
Nano	1	.6541	.4861	$-.1680$	$[-.2172, -.1209]$
Super	1	.5988	.3741	$-.2246$	$[-.2795, -.1742]$

Table 1: Planned benchmark comparisons: horizon-normalised own-payoff loss, 768 episodes per arm/temperature and all 48 paired blocks. Lower is better.

Opponent-specific results explain the average (Table 2). Draft improves all four means against AllC and Grim but worsens all four against TFT. Against unresponsive AllC, D maximises own payoff: improvement means more effective exploitation. TFT instead changes the value of a choice through its future response. At $\delta = .6$, optimal play alternates D,C; at .9, sustained cooperation is optimal. Draft's TFT reversal is concentrated in these regimes and the announced ten-round game; .4 improves in all four comparisons. The pattern cannot be explained by simply recommending more C or more D. Appendix A gives the incentive calculations.

Opponent	Nano 0	Super 0	Nano 1	Super 1
Always Cooperate	$-.5865$	$-.2094$	$-.6042$	$-.5167$
Always Defect	$+.0068$	$-.2083$	$-.1984$	$-.3385$
Grim	$-.0958$	$-.1511$	$-.1919$	$-.2151$
Tit-for-Tat	$+.2404$	$+.2103$	$+.3225$	$+.1717$

Table 2: Exploratory Draft–Direct loss differences by opponent. Each mean averages four conditions equally, with 192 episodes per arm/opponent/temperature.

3.2 Temperature, presentation and planning

Temperature has no uniform preferred setting. Super Direct improves at zero (0 minus 1: $-.1372$, nominal 95% interval $[-.1814, -.0942]$); Super Draft changes little ($-.0022$, $[-.0193, +.0147]$). Nano Direct also changes little ($-.0080$, $[-.0480, +.0316]$), whereas Nano Draft worsens ($+.0513$, $[+.0199, +.0815]$). The drafting advantage narrows at zero by $+.0592$ $[+.0099, +.1084]$ for Nano and $+.1350$ $[+.0876, +.1849]$ for Super.

At zero, Nano's exploratory drafting effect reverses with the neutral action mapping in both seats: $-.4652, -.2558$ when X means C, versus $+.1427, +.1431$ when X means D. The strata also use different stopping draws, motivating a fully crossed presentation experiment. Equal weighting retains both patterns in the primary estimate.

Against committed TFT/Grim with ten announced rounds, nine C moves followed by D earns the optimal 32 points. Only one of 768 episodes follows that path. Super Draft often makes the useful local TFT endgame choices (43/48 C moves at round nine and 48/48 D moves at round ten, at zero) while losing payoff earlier. This distinguishes a useful short plan from planning through the whole relationship.

3.3 Self-play separates cooperation from benchmark competence

Changing both self-play agents from Direct to Draft lowers episode-weighted mutual cooperation for each model at each temperature (Figure 1). Nano changes from 15.6% to 10.2% at zero and 24.6% to 5.1% at one; Super changes from 26.8% to 0.7% and from 13.9% to 2.3%, respectively. Each episode contributes its own CC fraction with equal weight. A pooled round rate weights longer episodes more; rounds within one trajectory share a history and are dependent.

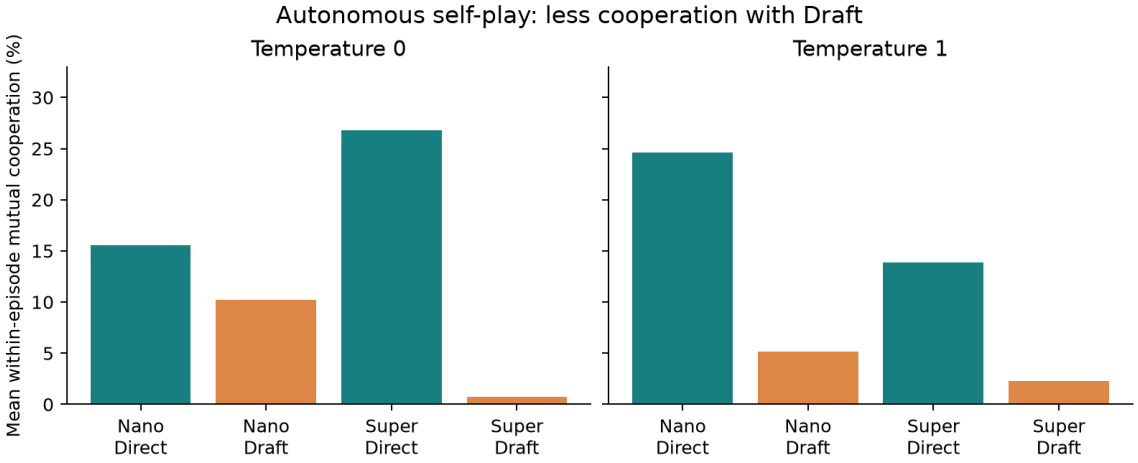


Figure 1: Mean within-episode mutual cooperation in autonomous self-play. Each comparison changes both agents; there are 48 dyads per pairing/temperature. These comparisons are exploratory.

Normalised joint return is $J = (1-\delta)\sum_t (r_{1,t} + r_{2,t})$, with $\delta = .9$. Its mean also falls under Draft in all four self-play comparisons: Nano 2.7500 → 2.5083 at zero and 3.3729 → 2.3854 at one; Super 3.2063 → 1.8667 and 3.1500 → 2.0792. This is a geometric-normalised episode total, rather than a realised per-round average. Each pairing's schedule contains 376 rounds over 48 dyads (7.833 rounds per dyad). For Super at zero, the joint-return difference is −1.3396, with exploratory nominal 95% paired-block interval $[-1.8375, -.9042]$.

Repair makes the contrast concrete. Super Direct at zero starts with no mutually cooperative first rounds, then returns to CC in 82/218 eligible transitions. Super Draft returns in 0/325 opportunities. The corresponding *round counts* are 124/376 and 3/376, with different denominators from the episode-weighted chart. Initial cooperation and the capacity to recover it are distinct behaviours.

4 Conclusions and future directions

A bounded draft improves average exact own-payoff decisions over the balanced benchmark, while responsive opponents and autonomous play reveal a different story for reciprocity. Local exploitation, contingent planning and cooperative repair emerge as separate evaluation targets. The interactive app connects these aggregate findings to individual decisions and histories, giving researchers, agent developers and students complementary ways to inspect them.

Planning through responses. Compare explicit state-transition plans at different depths, and vary draft length independently while matching compute. Score first actions and later continuation choices separately. Crossing every stopping seed with every label/seat setting would test whether the resulting plans remain stable under equivalent presentations.

Relationship formation and repair. Use controlled forgiving partners to vary when a cooperative probe or return to cooperation is possible. This would reveal how reciprocal exploration moves a pair into, maintains, or restores a productive interaction, and when individually defensive choices sustain low-cooperation patterns.

Communication and noise. Add messages alongside controlled action or observation errors. Compare commitments with subsequent choices and test whether explanations help agents distinguish mistakes from changed strategies. This links payoff reasoning, cooperation and commitment reliability in one protocol.

Longer and changing interactions. Separate available history from expected remaining interaction, and change a partner's rule during play. These interventions would distinguish memory and context use from incentives to preserve a relationship. Repeated encounters across episodes would make reputation and learning accessible to study.

Diverse incentives and populations. Vary temptation, shared surplus and punishment; compare asymmetric roles, larger groups and mixtures of simple policies, adaptive agents and humans. Partner selection would then connect the quality of individual decisions to which relationships and strategies persist across a population.

Evidence and reproducibility

Nebius Token Factory supplied hosted inference. Native Inspect AI samples, model events and custom scores [9], an append-only local event ledger, and MLflow Skinny/SQLite tracking [10] connect execution to analysis. All episodes completed; bounded recovery preserved the first 3,406 completed episodes and the original evidence prefix. Original and recovery source records remain separately frozen.

Public action histories reproduce exact scores, planned bootstrap intervals and exploratory diagnostics offline. The repository retains the full 16-condition and ten-pairing tables, numeric telemetry, dependency locks, source indices and verification commands. Private responses and draft text remain local. Resource use is measured from recorded request telemetry; usage-based cost estimates use the rates recorded at execution. Earlier development observations are excluded. Findings concern the stated prompts, hosted models and complete procedures; secondary and exploratory intervals have no family correction. Successful drafts reach their 1,024-token limit in 25–56% of cases, making bounded computation part of the observed intervention.

A Incentive calculations and app analysis

A.1 When future responses make cooperation pay

Against untriggered Grim, cooperation forever earns $R/(1-\delta)$; defecting now and best-responding to permanent D earns $T + \delta P/(1-\delta)$. Thus

$$\frac{R}{1-\delta} \geq T + \frac{\delta P}{1-\delta} \quad \Longleftrightarrow \quad \delta \geq \frac{T-R}{T-P} = \frac{1}{2}.$$

Against initially cooperating TFT, the three candidate continuation values are

$$\underbrace{\frac{3}{1-\delta}}_{\text{always C}}, \qquad \underbrace{5 + \frac{\delta}{1-\delta}}_{\text{always D}}, \qquad \underbrace{\frac{5}{1-\delta^2}}_{\text{alternate D,C}}.$$

Comparison makes initial C optimal for $\delta \geq 2/3$, explaining why .6 and .9 reward different patterns. With ten announced rounds against committed TFT/Grim, backward induction yields nine C moves and a final D, worth $9R + T = 32$. This uses a fixed response rule, rather than the different finite-game assumption in which both players remain free to optimise [1, Sections 4.1 and 5.1–5.2].

A.2 Reported forecasts in the educational app

Optional app forecasts are separate from the evaluation's action-only requests. Let p_t be a player's reported probability of the opponent's current C, and $F_{a,t}$ its expected own total *from the next round onward*, conditional on continuation and action a now. Define $\rho_t = \delta$ in a geometric game, 1 before an announced final round, and 0 at that final round. Then

$$\widehat{Q}_t(\mathrm{C}) = 3p_t + \rho_t F_{\mathrm{C},t},$$
$$\widehat{Q}_t(\mathrm{D}) = 1 + 4p_t + \rho_t F_{\mathrm{D},t},$$

so cooperation is weakly preferred according to those forecasts exactly when

$$\rho_t(F_{\mathrm{C},t} - F_{\mathrm{D},t}) \geq 1 + p_t.$$

The right side is the immediate temptation to defect; the left side is the forecast future advantage of C, weighted by continuation. Future totals lie between zero and $5/(1-\delta)$ in geometric games or $5(H-t)$ in fixed games. At a known final round, future advantage is zero and D pays more for every p_t. This checks consistency with submitted forecasts, not hidden beliefs; accuracy is assessed against outcomes actually observed. A one-step alternative holds the other player's sealed action fixed; longer alternatives need an explicit response model.

B Recreational games

Tic-tac-toe and Dots & Boxes are perfect-information mathematical games on grids or lattices, with visible sequential moves. They have their own scores and search references. For Tic-tac-toe, minimax from the current player's perspective has terminal values $+1, 0, -1$ and recursion $G(x) = \max_{a \in A(x)}[-G(f(x,a))]$.

Dots & Boxes grants another turn after a capture. If $g(x,a)$ is boxes captured and $W(x)$ is optimal *remaining*-box difference, then

$$W(x) = \max_{a \in A(x)} [g(x,a) + \sigma(x,a) W(f(x,a))], \qquad \sigma = \begin{cases} +1, & g(x,a) > 0, \\ -1, & g(x,a) = 0. \end{cases}$$

At completion $W = 0$; previously owned boxes are added separately. The sign preserves perspective on an extra turn and switches it when play passes. The app labels incomplete bounded searches unknown and presents chain/loop references only within their assumptions. These tactics complement the cooperation study in the separate Recreational games area.

References

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